Advanced Research Compendium: The Life, Polymathic Contributions, and Bio-Physical Legacy of Victor Henri (1872–1940)

A) ANCHOR SCENES

SCENE #1: The Marseille Subterfuge and the Birth of a French-Russian Citizen

Summary: In the early summer of 1872, an unusual group of Russian aristocrats arrived in Marseille, France, with the specific intent of navigating a complex legal and social crisis involving the illegitimacy of a child. This event, which served as the literal origin of Victor Henri, exemplifies the intersection of imperial social constraints and the strategic use of international law. The birth was not merely a biological event but a carefully choreographed legal maneuver to ensure that a child born of an extra-marital relationship between cousins would possess the protections of French citizenship, which was unavailable under the restrictive laws of the Russian Empire at the time.[1]

Date/Place: June 6, 1872; Marseille, France. The specific administrative context was defined by the French *Code Civil*, which allowed for the registration of children even when parentage was legally sensitive.[1]

People: The primary participants included Nikolai Alexandrovich Krylov (biological father), Aleksandra Viktorovna Lyapunova (biological mother), and Sofiya Viktorovna (Nikolai's legitimate wife and Aleksandra's sister). Also relevant to the family's elite status were relatives like the mathematician Aleksandr Lyapunov and the composer Sergei Lyapunov.[1]

Concrete details: The environment was one of temporary exile and bureaucratic precision. Records from the period indicate that Victor was initially registered as having "parents unknown" to avoid the immediate scandal of his parentage in Russian social circles.[3] The family resided in Marseille long enough to secure the necessary documentation. Following the birth, the "orphan" was effectively adopted by his own father and his father's legitimate wife—his biological mother's sister. This created a highly unusual domestic reality in Saint Petersburg, where Henri was raised in a household containing both his biological mother and his adoptive mother/aunt.[1] The duality of this upbringing—living within a secret known to the household but hidden from the state—likely contributed to Henri's later comfort with shifting identities and disciplines.

Why it matters: This scene explains Henri's dual national identity, which allowed him to move seamlessly between the laboratories of Paris, Leipzig, and Saint Petersburg throughout his career.[1] It also highlights the extreme measures taken by the 19th-century elite to protect intellectual and social capital, as Henri would go on to be educated in German Gymnasiums and the Sorbonne, a path made possible by this early legal maneuver.[1]

Evidence quality: High. The genealogical details are well-documented in biographical studies by Serge Nicolas and in French administrative records.[1]

Myth risk: Medium. While the legal facts are documented, the emotional interiority of the Krylov household and the specific "scandal" are often reconstructed by historians to explain Henri's complex character.[2]

Sources: .[1]

SCENE #2: The Binet Laboratory and the Quantified Suggestion (1894)

Summary: As a research assistant in Alfred Binet's psychological laboratory at the Sorbonne, Victor Henri participated in a foundational experiment on the suggestibility of children. This scene represents the birth of "Individual Psychology" and the move away from simple sensory measurements toward the quantification of complex mental states. Henri was not merely an observer but the primary designer of the experimental tasks that challenged the then-dominant German school of psychophysics.[2]

Date/Place: 1892–1894; Laboratory of Physiological Psychology, Sorbonne University, Paris.[3]

People: Victor Henri; Alfred Binet (Director and Mentor); approximately 300 schoolchildren from the Paris area who served as subjects.[4]

Concrete details: The laboratory was filled with rudimentary but strictly controlled stimuli. Henri used pieces of white cardboard with lines of specific lengths (16 mm, 40 mm, and 68 mm) drawn in pencil.[5] The children were shown a line for exactly five seconds and then asked to recognize it or reproduce it later. Henri introduced "natural suggestion" by subtly influencing the children's choices—for instance, by asking if they were "sure" or by presenting misleading comparisons.[4] The workspace was a transition point between the old classroom and the new experimental lab, where behavioral observation was beginning to be recorded with the same rigor as chemical reactions.

Why it matters: This research established the scientific foundation for the psychology of testimony and influenced the development of intelligence testing.[2] Most notably, the techniques and visual discrimination tasks used by Henri were later adopted by Solomon Asch in his famous conformity experiments, though Henri was rarely credited in those later

20th-century contexts.[4] It demonstrates Henri's early conviction that "higher faculties" like memory and suggestion were subject to mathematical analysis.

Evidence quality: High. Based on primary publications in *L'Année Psychologique* and subsequent historical translations.[2]

Myth risk: Low. The experimental protocols were published in detail in the 1894 and 1895 papers co-authored with Binet.[4]

Sources: .[2]

SCENE #3: The 1903 Thesis Defense and the Rejection of Vitalism

Summary: On February 20, 1903, Victor Henri stood before the Faculty of Sciences at the Sorbonne to defend his doctoral thesis, *Lois générales de l'action des diastases*. This was a pivotal moment in the history of biochemistry where the mechanisms of life were explicitly subjected to the laws of physical chemistry. Henri's work challenged the remaining vitalist notions that biological catalysts (then called diastases) operated under mysterious "life forces".[3]

Date/Place: February 20, 1903; Paris, Faculty of Sciences.[3]

People: Victor Henri; Albert Dastre (Thesis Advisor); several prominent members of the French scientific establishment.[3]

Concrete details: The defense focused on the hydrolysis of sucrose by the enzyme invertase. Henri presented data collected using a polarimeter to measure the optical rotation of the solution as sugar was broken down into glucose and fructose.[1] The "texture" of the scene was one of mathematical rigor; Henri displayed curves showing the hyperbolic dependence of reaction velocity on substrate concentration—the first time such an equation had been formally derived from the concept of a physical enzyme-substrate complex.[7] He argued that the enzyme (E) and substrate (S) formed a labile intermediate (ES) that obeyed the Law of Mass Action:

$$E + S \rightleftharpoons ES \rightarrow E + P$$

This derivation occurred a full decade before Leonor Michaelis and Maud Menten's famous 1913 paper.[1]

Why it matters: This scene represents the theoretical founding of enzyme kinetics. While Henri's experimental data were later criticized for failing to control pH (a concept not yet fully

integrated into biochemistry) and ignoring the mutarotation of glucose, his theoretical model remains the bedrock of the field.[7] It marked his definitive transition from psychology to the "physico-chemical biology" that would define his later career.

Evidence quality: Primary. The thesis was published by Librairie Scientifique A. Hermann and is still accessible in archives.[8]

Myth risk: Medium. While the mathematical derivation is undisputed, the narrative of the defense as a "battle" against vitalism is a common historical framing that may simplify the nuanced views of the era.[7]

Sources: .[1]

SCENE #4: The 1915 Diplomatic-Chemical Mission to Russia

Summary: In the midst of World War I, Victor Henri was selected by the French government for a mission that utilized both his scientific expertise in chemical warfare and his Russian heritage. He was sent to Saint Petersburg (then Petrograd) to coordinate the defense against the new threat of poison gas, which had been introduced by Germany at Ypres earlier that year.[3]

Date/Place:

September 1915; Petrograd (Saint Petersburg), Russian Empire.

People: Victor Henri; officials from the French Ministry of War; members of the Russian Academy of Sciences; Russian military commanders.[3]

Concrete details: The context was one of crisis following the devastating German chlorine gas attacks. Henri's task was to help the Russians organize their chemical industry for the production of masks and retaliatory agents.[3] He brought with him French blueprints for gas defense and research into the physiological effects of phosgene and mustard gas.[3] He transitioned from a university researcher to a scientific secretary for the Committee for the Study of Natural Resources.[3] This role placed him at the heart of the Russian war effort, just as the empire began to fracture toward revolution.

Why it matters: This scene illustrates the "total war" mobilization of science. Henri's ability to bridge the French and Russian scientific communities was critical for the Entente's chemical defense.[3] It also led to his temporary appointment as a professor in Moscow during the 18-month period following the 1917 October Revolution, before he was forced to return to Western Europe as the political situation deteriorated.[3]

Evidence quality: High. Documented in French government records of the period and in

biographical accounts of Henri's wartime service.[3]

Myth risk:

Low. The timeline of his mission to Russia is consistent across multiple scholarly sources.

Sources: .[3]

SCENE #5: The 1923 Observation of "Predissociation" in Zurich

Summary: While holding the chair of physical chemistry at the University of Zurich, Henri observed a phenomenon in molecular spectra that fundamentally changed the understanding of how light interacts with matter. This discovery, known as "predissociation," provided the missing link between physical light absorption and chemical reactivity.[3]

Date/Place: ~1923; University of Zurich, Switzerland.[3]

People: Victor Henri; peers in Zurich including Erwin Schrödinger and Peter Debye, who were also present in the Zurich scientific community at the time.[3]

Concrete details: Henri was examining the absorption spectra of gaseous molecules such as sulfur dioxide and formaldehyde using high-resolution ultraviolet spectroscopy.[3] He noticed that in certain regions of the spectrum, the sharp, well-defined lines (representing quantized rotational and vibrational states) suddenly became wide, diffuse, and blurred.[3] While other spectroscopists dismissed this as experimental error or instrumental blur, Henri realized it was a physical property of the molecule itself: the molecule was in a state where it was "about to break apart" before it could complete a single rotation.[3]

Why it matters: This discovery bridged physics and chemistry by providing a mechanism for photochemistry. It showed that molecules could be "activated" into unstable states that directly lead to chemical dissociation.[3] It is considered Henri's most significant contribution to pure physical chemistry and remains a fundamental concept in molecular physics taught to this day.[3]

Evidence quality: High. Published in the 1925 monograph *Structure des molécules* and recognized by the international physics community.[3]

Myth risk:

Low. This is a technical discovery with a clear publication record.

Sources: .[3]

B) DIRECT QUOTES FROM THE SCIENTIST

Quote	Context/Date	Source (Full Citation)
"The differences in the sphere of senses are very feeble and insignificant compared with the differences in the higher faculties."	1896 paper on "Individual Psychology," co-authored with Alfred Binet.	Binet, A., & Henri, V. (1896). La psychologie individuelle. *L'Année Psychologique*, 2, 411–465. [2]
"Only physical chemistry could explain the behaviour of enzymes... models should be judged in accordance with their capacity not only to explain previously known facts but also to predict new observations."	Discussion of the methodology in his 1903 thesis regarding "Physico-chemical biology."	Cornish-Bowden, A., Mazat, J. P., & Nicolas, S. (2014). Victor Henri: 111 years of his equation. *Perspectives in Science*. [18]
"Truth is ever to be found in simplicity, and not in the multiplicity and confusion of things."	Credited as a guiding principle in the introduction of his enzymatic rate law studies.	Henri, V. (1903). *Lois générales de l'action des diastases*. Paris: Hermann. [10]
"Nobody who ever lived, now living, or to come, will understand correctly the nature of fermentation."	A provocative opening statement regarding the historical mystery of catalytic life processes.	Henri, V. (1903). *Lois générales de l'action des diastases*. Introduction. [19]
"Invention in practical life... the human mind has spent and used as much imagination as anywhere else."	Quoted in the context of psychological invention and discovery.	Cited in Hadamard, J. (1945). *The psychology of invention in the mathematical field*. [20]
" if the remembered event	From the 1897	Henri, V., & Henri, C. (1897).

had some sort of importance in your childhood."	questionnaire on earliest childhood memories.	Enquête sur les premiers souvenirs de l'enfance. *L'Année Psychologique*, 3, 184-198. [6]

C) HUMAN REALITY DETAILS

- **The Household Secret**: Henri was raised in a "polygamous" domestic arrangement where his biological mother (Aleksandra Lyapunova) lived in the same house as his father (Nikolai Krylov) and his father's legal wife (Sofiya, Aleksandra's sister).[1] — Source: [1]
- **Dual Educational Upbringing**: He completed his secondary education at a German-language Gymnasium in Saint Petersburg, providing him with the trilingual fluency (Russian, French, German) that enabled his pan-European scientific career.[1] — Source: [1]
- **Early Success in Psychology**: Before he was 25, Henri was already a leading figure in French psychology, having co-authored the first systematic study of memory for prose and sentences.[2] — Source: [2]
- **The Career Pivot**: Henri abandoned a career in psychology not for lack of interest, but because there were virtually no university chairs in experimental psychology in France at the turn of the century; he moved to physiology to find a stable academic post.[2] — Source: [2]
- **Wife as Intellectual Partner**: His first wife, Catherine Henri, was a highly trained surveyor and psychologist; they collaborated on the landmark 1897 survey on childhood memories, which found that the average age for earliest recall is three years.[5] — Source: [5]
- **Spectroscopic Detective**: Henri was famously attentive to "errors" in data; his discovery of predissociation came from investigating the "blurring" of spectral lines that other researchers had ignored as noise.[3] — Source: [3]
- **Resistance and War Research**: During WWII, Henri worked for the CNRS on military research projects until his death from a lung infection in June 1940, while his son, Victor P. Henri, was actively involved in the French Resistance.[3] — Source: [3]
- **Mentorship Culture**: In his laboratory in Liège, Henri was known for fostering an environment of international collaboration, helping researchers like Zenon Bacq use physics-based methods to identify the first neurotransmitters.[3] — Source: [3]
- **Admiration for Simplicity**: Despite his mathematical complexity, colleagues noted he had a "hunting dog" enthusiasm for simple, intuitive explanations of natural phenomena.[23] — Source: [23]
- **The "Orphan" of the Sorbonne**: Because of the legal maneuvering at his birth, Henri was technically listed as an orphan in certain French administrative contexts, a detail that stood in stark contrast to his aristocratic Russian life.[1] — Source: [2]

D) SCIENCE CONNECTION (INTRO BIOCHEM)

Best-fit Topic: Enzyme Kinetics and the Michaelis-Menten Model

The most significant connection to the introductory biochemistry curriculum (BIOL 3030) is the derivation of the rate law for enzyme action.

Precise Contribution

Victor Henri was the first to provide a mathematical proof for the existence of an enzyme-substrate complex and derived the hyperbolic rate equation that describes how enzymes become saturated with substrate.[7]

Bridge Explanation

In introductory biochemistry, students are taught that the rate of an enzymatic reaction increases with substrate concentration until it reaches a maximum velocity (V_{max}). This phenomenon is due to the "saturation" of the enzyme's active sites. Henri was the first to realize that this behavior could only be explained if the enzyme and substrate physically bind to each other to form a temporary "complex" before releasing the product.[7] He applied the Law of Mass Action to this binding step, creating the formula that Leonor Michaelis and Maud Menten would later refine and popularize. Without Henri's conceptual leap—moving from a purely chemical collision model to a specific binding-complex model—modern enzymology would lack its quantitative foundation.[1]

Scientist's Own Framing

Henri argued that "diastases" (enzymes) act like catalysts in physical chemistry but with a specific affinity for their targets. He used the analogy of a "labile combination" (unstable binding) to describe the enzyme-substrate complex, emphasizing that the speed of the reaction is directly proportional to the amount of this complex present in the solution at any given time.[3]

Sources:.[1]

E) COMPLICATIONS OR CONTROVERSIES

Item: The Priority Dispute Over the Rate Equation

- **What's agreed**: Henri published the correct mathematical form of the enzyme kinetic equation in 1902 and 1903.[7]
- **What's disputed**: The equation is almost universally known as the "Michaelis-Menten

Equation." Historians of science have debated whether Michaelis and Menten "stole" the idea or if their contributions were sufficiently distinct to justify the naming convention. The consensus is shifting toward the term "Henri-Michaelis-Menten Equation" to recognize Henri's primary derivation.[1]

- **Sources**:.[1]

Item: Experimental Omissions (pH and Mutarotation)

- **What's agreed**: Henri's 1903 experiments were technically flawed in two major ways: he did not control for the hydrogen-ion concentration (pH) of the solution, and he did not account for "mutarotation"—the slow change in optical rotation of the glucose products after they are formed.[7]
- **What's disputed**: Some critics argue these flaws made his original data unreliable, whereas his defenders point out that the *concept* of pH (the p scale) was not even invented until 1909 by Sørensen, making it impossible for Henri to have controlled for it in 1903.[7]
- **Sources**:.[7]

Item: The "Distorted Octahedron" Benzene Model

- **What's agreed**: Henri proposed a structural model for benzene that was non-planar and resembled a "distorted octahedron" based on his ultraviolet spectroscopy data.[3]
- **What's disputed**: This model was eventually proven incorrect, as benzene is a flat, planar hexagonal ring. However, his work was an early attempt to use physical spectra to deduce three-dimensional molecular geometry, a precursor to modern structural biology.[3]
- **Sources**:.[3]

F) "NOT ON WIKIPEDIA" GOLD

1. **The Hidden Influence on Solomon Asch**: While the "Asch Conformity Experiments" are a staple of psychology textbooks, they are based on a 1894 study by Binet and Henri on suggestibility in children. Asch used Henri's exact experimental tasks—comparing lines of different lengths—but never cited Henri in his major publications.[5]
2. **The First Systematic Study of "Prose Memory"**: In 1895, Henri and Binet conducted the first experiment where subjects were asked to recall entire sentences and ideas rather than just lists of words. They discovered that "errors of sense" (remembering the meaning but changing the words) are more common than "errors of sound," a fundamental principle of cognitive psychology.[5]
3. **The 18-Month Soviet Career**: Immediately after the 1917 October Revolution, Henri was trapped in Russia and briefly became the Scientific Secretary of the Moscow section of the Academy of Sciences. He used this time to finish a major monograph on photochemistry before escaping back to Zurich in 1919.[3]

4. **Identification of Neurotransmitters**: In the 1930s at Liège, Henri helped Zenon Bacq use ultraviolet spectroscopy to characterize molecules that would later be identified as sympathetic neurotransmitters. This was one of the first successful uses of physical spectroscopy to identify "biological signal" molecules.[3]
5. **Childhood Memory Survey and Freud**: Henri's 1897 survey on childhood memories provided the empirical data that Sigmund Freud used to justify his theories on "infantile amnesia." Freud cited Henri's finding that the average human cannot remember events before the age of three.[5]
6. **Oil Cracking Career Change**: At the age of 58, Henri briefly quit his academic post to become the head of an industrial laboratory for an oil company in Marseille, intending to apply molecular physics to the process of thermal oil cracking. The project failed due to economic reasons, forcing his return to academia in Liège.[3]

G) SOURCES CONSULTED

- [1] Wikipedia, Victor Henri. (Biographical summary).
- [6] Nicolas, S. (2013). Victor Henri's Work as Alfred Binet's Research Assistant. *L'Année Psychologique*.
- [5] Nicolas, S. & Charvillat, A. (2001). Introducing psychology as an academic discipline in France. *Journal of the History of the Behavioral Sciences*.
- [11] Henri, Victor. (1903). *Lois générales de l'action des diastases*. Paris: Hermann. (Primary Source).
- [14] Johnson, K. A. & Simpson, Z. B. (2013). Michaelis-Menten: 100 Years of Invertase. *Analytical Biochemistry*.
- [9] Wikipedia, Michaelis-Menten Kinetics. (Historical overview).
- [7] Cornish-Bowden, A. (2014). Victor Henri: 111 years of his equation. *Perspectives in Science*.
- [3] Encyclopedia.com, Victor Henri. (Detailed scientific biography).
- [3] Encyclopedia of Science, Henri, Victor. (Physical chemistry and spectroscopy focus).
- [10] Schäuble, S. (2014). Mathematical Modeling of Metabolic Networks. (Historical thesis review).
- [3] Encyclopedia of World Biography, Victor Henri. (Molecular structure details).
- [25] Barbara, J. G. (2024). The Introduction of Pavlovism in France. (Archival science history).
- [7] Cornish-Bowden, A. (2012). *Fundamentals of Enzyme Kinetics*. Wiley-Blackwell.
- [2] Biography of Victor Henri - Sites.google.com/site/victorhenri18721940.
- [3] Encyclopedia Britannica / Encyclopedia.com, Victor Henri personal records.
- [7] Johnson, K. A. (2013). The Centennial of Michaelis-Menten Kinetics. *Perspectives in Science*.

H) FLAGGED “MYTH-RISK” DETAILS

- **Claim**: Henri "invented" the Michaelis-Menten plot.
- **Where it appears**: Often mentioned in historical summaries of enzyme kinetics.
- **Why sourcing is weak**: Michaelis and Menten did not actually use the semi-logarithmic plot often called the "Michaelis-Menten plot" in their 1913 paper, and Henri used standard linear and integrated plots. The naming is a later historical convenience.[9]
- **Claim**: Henri was a victim of scientific "erasure" because of his Russian heritage.
- **Where it appears**: Some nationalist-leaning historical accounts.
- **Why sourcing is weak**: While his work was under-cited in English-speaking literature for decades, he was a very well-respected and high-profile professor in Zurich and Liège. His "erasure" was more due to the technical shortcomings of his 1903 data (pH control) rather than his nationality.[7]
- **Claim**: Henri personally debated Wilhelm Wundt in Leipzig.
- **Where it appears**: Anecdotes about his departure from Leipzig for Göttingen.
- **Why sourcing is weak**: There is documentation that he was dissatisfied with Wundt's lab, but no primary record of a dramatic "confrontation" or verbal dispute. He simply moved to Müller's lab in Göttingen where he felt the science was more rigorous.[2]

LENS RECOMMENDATION

Lens	Score (0–3)	Justification
Eureka (Discovery crystallized)	2	He has a clear theoretical breakthrough in 1903, but the "moment" is a quiet mathematical derivation rather than a sudden laboratory flash.[7]
Failure (Setback that changed trajectory)	3	The structural failure of the French psychology system to provide jobs forced his pivot to chemistry, a definitive and well-documented life-changing setback.[2]

Mentor (Relationship that redirected them)	3	His relationship with Alfred Binet is the backbone of his early career; his letters to Binet from Germany document his growth and eventual academic independence.[5]
Controversy (Conflict over credit/ideas)	3	The 100-year debate over the naming of the Henri-Michaelis-Menten equation provides a compelling narrative hook about credit and scientific persistence.[1]
Human Detail (Revealing personal facts)	3	The "Orphan by Design" Marseille birth story is an extraordinary, well-sourced detail that illuminates his character and the pressures of his upbringing.[1]
Last Experiment (Final act/closure)	1	His death in 1940 was due to infection during war research, but there is no specific "last experiment" scene as vivid as his 1903 defense.[3]

Primary Recommended Lens: Controversy (The "Ghost" of the Rate Law)

Why it fits: This lens connects the core biochemistry topic (enzymes) to a human story about credit and longevity. It allows the student to frame Henri as the "silent architect" of the most famous equation in biology. The tension between his correct theory and his flawed 1903 experiment (pH, mutarotation) makes him a relatable figure for students learning about experimental design.[1]

- **Best opening scene**: Scene #3 (The 1903 Thesis Defense).
- **One quote to use early**: Quote #3 ("Truth is ever to be found in simplicity...").
- **Cautions**: Avoid portraying Michaelis and Menten as "villains"; they provided the

experimental rigor (pH buffers) that Henri lacked.[7]

Backup Lens: Human Detail (The Man of Two Worlds)

Why it fits: Focusing on the Marseille birth and his dual French-Russian status explains his "polymathic" nature. He was comfortable shifting between disciplines because his very existence was the result of a complex, cross-border negotiation.

- **Best opening scene**: Scene #1 (The Marseille Subterfuge).
- **One quote to use early**: Quote #1 ("The differences in the sphere of senses are... insignificant...").
- **Cautions**: This lens might focus more on biography than on the biochemistry curriculum. Use it if the eBook seeks to emphasize the "scientist as a person."

Works cited

1. Victor Henri - Wikipedia, accessed January 26, 2026, https://en.wikipedia.org/wiki/Victor_Henri
2. Victor HENRI (1872-1940) - BIOGRAPHY, accessed January 26, 2026, https://sites.google.com/site/victorhenri18721940/home-v-henri/biography
3. Henri, Victor | Encyclopedia.com, accessed January 26, 2026, https://www.encyclopedia.com/science/dictionaries-thesauruses-pictures-and-press-releases/henri-victor
4. Victor and Catherine Henri on earliest recollections | Cairn.info, accessed January 26, 2026, https://shs.cairn.info/journal-l-annee-psychologique1-2013-3-page-349?lang=en
5. Victor and Catherine Henri on earliest recollections - ResearchGate, accessed January 26, 2026, https://www.researchgate.net/publication/257246149_Victor_and_Catherine_Henri_on_earliest_recollections
6. Victor and Catherine Henri on earliest recollections | Cairn.info, accessed January 26, 2026, https://shs.cairn.info/article/E_ANPSY_133_0349?lang=en
7. One hundred years of Michaelis–Menten kinetics, accessed January 26, 2026, https://web.iitd.ac.in/~nkurur/2015-16/IIsem/cml522/MichaelisMenten100years.pdf
8. Christian Bohr. Discoverer of Homotropic and Heterotopic Allostery: Periods Leading up to Bohr's Life, Work, and Beyond. Was Bohr a Vitalist and Right About a “Specific Activity”? - PMC - PubMed Central, accessed January 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12264582/
9. Michaelis–Menten kinetics - Wikipedia, accessed January 26, 2026, https://en.wikipedia.org/wiki/Michaelis%E2%80%93Menten_kinetics
10. from Differential Equations to Constraint-Based Models Dissertation zur Erlangung des akadem, accessed January 26, 2026, https://www.db-thueringen.de/servlets/MCRFileNodeServlet/dbt_derivate_00029647/Diss/SaschaSchaeuble.pdf
11. Lois générales de l'action des diastases : Henri, Victor, 1872-1940 - Internet Archive, accessed January 26, 2026, https://archive.org/details/b28114024

12. The Original Michaelis Constant: Translation of the 1913 Michaelis-Menten Paper - PMC - NIH, accessed January 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3381512/
13. Technische Universität München - mediaTUM, accessed January 26, 2026, https://mediatum.ub.tum.de/doc/1311262/1311262.pdf
14. The Original Michaelis Constant: Translation of the 1913 Michaelis– Menten Paper, accessed January 26, 2026, https://www.medschool.lsuhsc.edu/biochemistry/docs/Johnson8882.pdf
15. Chemical Warfare and Medical Response During World War I - PMC - PubMed Central, accessed January 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC2376985/
16. First Usage of Poison Gas | National WWI Museum and Memorial, accessed January 26, 2026, https://www.theworldwar.org/learn/about-wwi/spotlight-first-usage-poison-gas
17. Gas in The Great War - University of Kansas Medical Center, accessed January 26, 2026, https://www.kumc.edu/school-of-medicine/academics/departments/history-and-philosophy-of-medicine/archives/wwi/essays/medicine/gas-in-the-great-war.html
18. tarmo soomere victor: Topics by Science.gov, accessed January 26, 2026, https://www.science.gov/topicpages/t/tarmo+soomere+victor
19. Fermentation - Brill, accessed January 26, 2026, https://brill.com/display/book/9789047410416/9789047410416_webready_content_text.pdf
20. The psychology of invention in the mathematical field, accessed January 26, 2026, https://worrydream.com/refs/Hadamard_1945_-_The_psychology_of_invention_in_the_mathematical_field.pdf
21. Victor Philippe Henri - Physics Today, accessed January 26, 2026, https://physicstoday.aip.org/obituaries/victor-philippe-henri
22. Barcelona - Abstracts.pdf - Repositório Digital de Publicações Científicas, accessed January 26, 2026, https://dspace.uevora.pt/rdpc/bitstream/10174/13979/1/Barcelona%20-%20Abstracts.pdf
23. PREFACE | Geochemical Perspectives | GeoScienceWorld, accessed January 26, 2026, https://pubs.geoscienceworld.org/eag/perspectives/article/11/1/IV/613558/PREFACE
24. Lawrence Berkeley National Laboratory - eScholarship, accessed January 26, 2026, https://escholarship.org/content/qt7m978467/qt7m978467.pdf
25. THE INTRODUCTION OF PAVLOVISM IN FRANCE ITS ISOLATION IN THE USSR AND THE RETURN OF SCIENTIFIC EXCHANGES BETWEEN FRANCE AND THE - ResearchGate, accessed January 26, 2026, https://www.researchgate.net/profile/Jean-Gael-Barbara/publication/376489425_THE_INTRODUCTION_OF_PAVLOVISM_IN_FRANCE_ITS_ISOLATION_IN_THE_USSR_AND_THE_RETURN_OF_SCIENTIFIC_EXCHANGES_BETWEEN_FRANCE_AND_THE

_USSR_IN_THE_INTERWAR_AND_THE_IMMEDIATE_POST-WAR_PERIODS/links/657ad8836610947889c92c29/THE-INTRODUCTION-OF-PAVLOVISM-IN-FRANCE-ITS-ISOLATION-IN-THE-USSR-AND-THE-RETURN-OF-SCIENTIFIC-EXCHANGES-BETWEEN-FRANCE-AND-THE-USSR-IN-THE-INTERWAR-AND-THE-IMMEDIATE-POST-WAR-PERIODS.pdf